

## **Design Optimization Proposal for Enhanced Robotic Arm Responsiveness**

### **1. Problem Statement**

There's a delay in the rotate\_joint command for model number-2201. Investigation showed a 50% increase over the expected time for the rotate\_joint command which impacts real-time responsiveness in surgical applications.

#### **Identified Issues:**

##### **1. Actuator Strain:**

- **Description:**  
The actuator is worn down from repeated use. This is likely due to the materials being used and the stress caused on the actuators over time.
- **Impact:** [Explain how this issue impacts performance]  
This negatively impacts the precision and reliability of the model's arm while decreasing its strength continuously.

##### **1. Sensor Misalignment:**

- **Description:**  
The sensors are either not placed in the most strategically effective positions or theirs a lack of recalibration.
- **Impact:**

This provides inaccurate data to the processing unit, which can lead to miscalculations in the control code. Inaccurate information will negatively impact the sensor's ability to maintain surgical precision over long periods of time.

### **2. Proposed Modifications**

#### **Modification 1:**

- **Proposal:**  
Carbon fiber or advanced alloys should be used to repair and boost joint durability.
- **Justification:**  
The recommended materials are both lightweight and strong, so it provides additional durability without sacrificing weight or precision.

#### **Modification 2:**

- **Proposal:** [Describe the modification]

Improve processes to regularly calibrate and strategically place sensors effectively. Any project management tools or tracking tools are recommended to keep track of both sensor placements and team member responsibility.

- **Justification:**

The changes provide a more reliable source of information that comes from sensors once they're consistently aligned while strategic placement will always be improved upon.

### 3. Simulation Results

- **Initial Metrics:**

- **Actuator Response Time: .20 seconds**
- **Sensor Feedback Delay: 75/100**

- **Post-Optimization Metrics:**

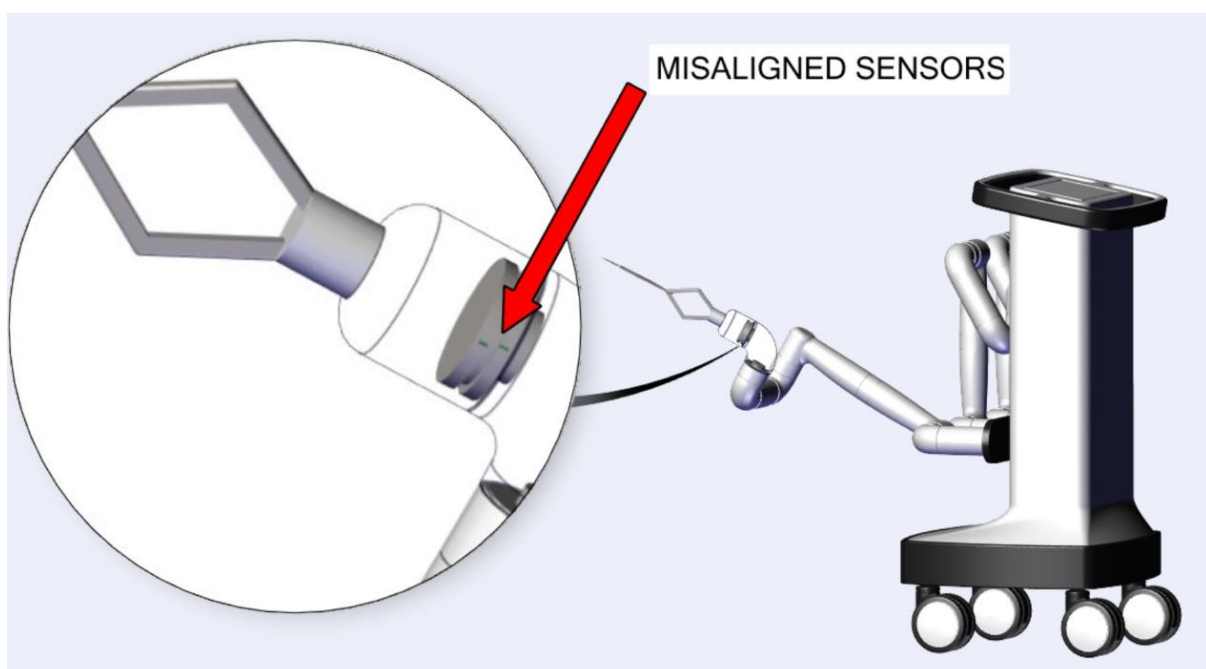
- **Actuator Response Time: .16**
- **Sensor Feedback Delay: 60/100**

- **Key Insights:**

- Actuator response time is faster due to the new materials added to the actuator
- Sensor feedback delay decreases misalignment and better placement in a single run.

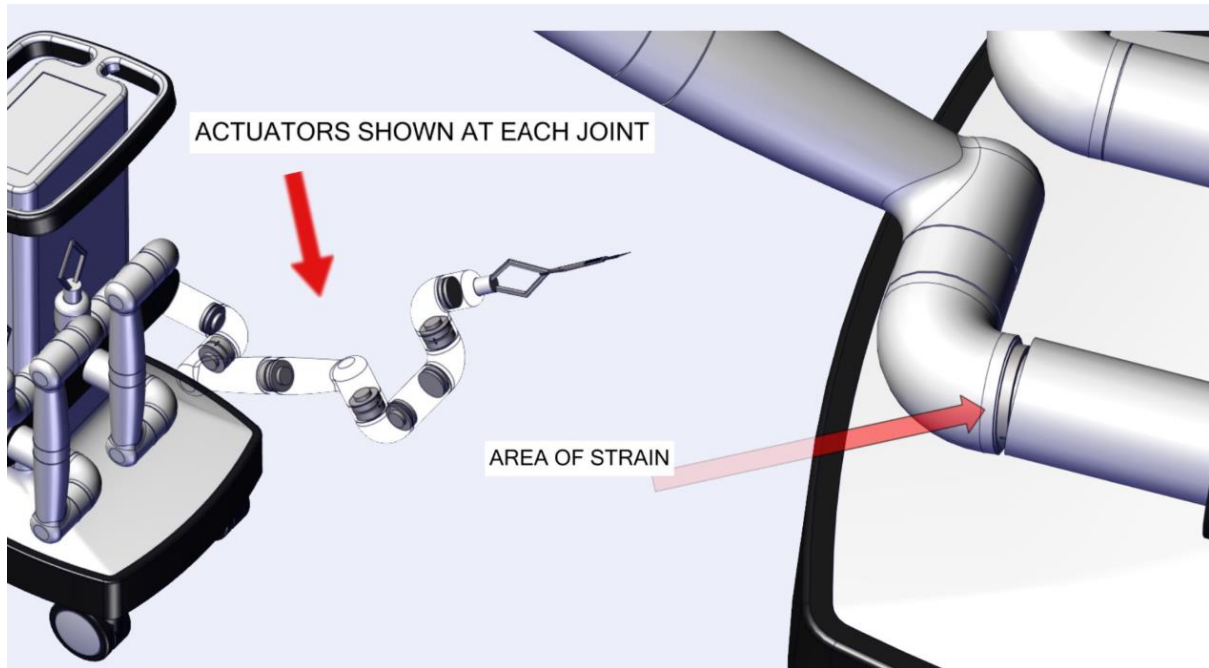
### 4. Annotated Visual Examples

#### Visual Example 1: Current Sensor Misalignment vs. Proposed Realignment



- *Brief Description:* The two green lines not being on top of each other represent the misalignment in the sensors.

### Visual Example 2: Actuator Upgrade Impact



- *Brief Description:* The actuators are placed within the joins of the arm and can potentially tear or wear down over time due to frequent use.

## 5. Expected Outcomes

### 1. Improved Responsiveness:

Providing stronger material to the actuators along with more accurate information from sensors improves responsiveness from the processing unit by giving it better data with less room for error.

### 2. Enhanced Reliability:

Providing consistently accurate information from sensors allows the processing unit to deal with fewer edge cases or potentially deal with errors. Improving the strength and durability of the actuator's lengths at the time an arm can be used without repair.

### 3. Increased Durability:

Providing stronger but lightweight materials that don't sacrifice precision will greatly improve the durability of the actuators by putting less stress on the actuator and arm.

## 6. Conclusion

Summarize the impact of your proposed modifications:

The proposed modifications will directly impact both key components it seeks to address while also indirectly boosting productivity for the processing unit.